

控制理论中的代数基础试题(A)

一、Justify each answer: draw “√” if True and “×” if False. (30 points)

1. The subset $S = \{(w_1, w_2) \in \mathbb{R}^2 \mid w_1 w_2 \leq 0\}$ is a subspace of $\mathbb{R}^2$. ()
2. $\mathbb{R}^2$ is a subspace of $\mathbb{R}^5$. ()
3. If U is a subspace of a finite-dimensional vector space V , then $\dim(U) \leq \dim(V)$. ()
4. Let $\{v_1, \dots, v_p\}$ be a spanning set of V . If $\dim V = p$, then $\{v_1, \dots, v_p\}$ is a basis of V . ()
5. Every finite-dimensional vector space has a unique basis. ()
6. For a linear transformation $T \in \mathcal{L}(V, W)$, if $v_1, \dots, v_p$ is linearly independent in V , then $T(v_1), \dots, T(v_p)$ is also linearly independent in W . ()
7. Let $V = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \right\}$ and $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$. The matrix representing a linear transformation from V to W is a 2×3 matrix. ()
8. A linear transformation to a higher-dimensional space can be injective. ()
9. If $e_1, e_2, \dots, e_m$ is an orthonormal list and v is any vector, then $v - \sum_{j=1}^m \langle v, e_j \rangle e_j$ is orthogonal to each e_j . ()
10. If u and v are nonzero vectors in an inner product space and $\|u + v\| = \|u\| + \|v\|$, then u and v are scalar multiples of each other. ()
11. If T has an upper-triangular matrix whose diagonal entries are all distinct, then T is diagonalizable. ()
12. If every vector in a basis of a linear space V is an eigenvector of T , then every nonzero vector in V is an eigenvector of T . ()
13. If V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$, then T has an upper-triangular matrix with respect to some basis of V . ()
14. If W is a subset, then W and $W^\perp$ may have no vectors in common. ()
15. In a real inner product space, if two nonzero vectors u and v are linearly dependent, then they must be orthogonal. ()

二、(4 points)

Let W be a finite-dimensional vector space, and let U and V be two subspaces of W . Suppose that $U \cap V = \{0\}$, prove that $\dim(U + V) = \dim U + \dim V$.

三、(6 points)

$S = \{(x_1, x_2, x_3, x_4) : x_1 - 4x_2 + 3x_4 = 0\}$ is a subspace of $\mathbb{R}^4$.

(a) (3 points) Write a basis of S .

(b) (3 points) Write a basis of a subspace W of $\mathbb{R}^4$ satisfying $\mathbb{R}^4 = S \oplus W$.

四、(10 points)

Let $A \in \mathbb{R}^{3 \times 4}$ with reduced row echelon form give by $R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The first

column of A is $a_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $x_0 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is a solution of $Ax_0 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$.

(a) (4 points) Find bases for the four fundamental subspaces of A .

(b) (6 points) Write down the fundamental theorem of linear algebra: Part 1 and Part 2, and verify them using the above four fundamental subspaces.

五、(8 points)

Compute.

(a) (4 points) For the linear transformation $T(x) = \begin{bmatrix} x_1 - 2x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix}$ with $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ from $\mathbb{R}^2$ to

$\mathbb{R}^3$, find the matrix representing T with respect to the ordered basis $E = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}$

and $F = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \right\}$.

(b) (4 points) Let U be the subspace of $C[a, b]$ spanned by e^x, xe^x, x^2e^x . Let D be the differentiation operator on U . Find the matrix A representing D with respect to the ordered basis $[(1-x)e^x, -e^x, (x^2-x)e^x]$.

六、(10 points)

Suppose that V and W are finite-dimensional vector spaces and $T_1, T_2 \in \mathcal{L}(V, W)$.

- (a) (4 points) For $T \in \mathcal{L}(V, W)$, write down the fundamental theorem of linear maps.
- (b) (3 points) Suppose that U is a three-dimensional subspace of $\mathbb{R}^8$ and that T is a linear map from $\mathbb{R}^8$ to $\mathbb{R}^5$ such that $\text{null } T = U$. Use the above theorem to prove that T is surjective.
- (c) (3 points) If $\text{range } T_1 \subset \text{range } T_2$, show that there is a linear operator $S \in \mathcal{L}(V)$ such that $T_1 = T_2 S$.

七、(8 points)

Let $T \in \mathcal{L}(\mathbb{C}^3)$ have matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

with respect to the standard basis.

- (a) (4 points) Find the only eigenvalue of T and its eigenspace.
- (b) (4 points) Determine whether T is diagonalizable.

八、(8 points)

In $\mathbb{R}^4$ with the standard inner product, let

$$U = \text{span}\{(1, 1, 0, 0), (1, 0, 1, 0), (1, 0, 0, 1)\}$$

and let $v = (1, 2, 3, 4)$. Find the vector in U closest to v , and find the distance from v to U .

九、(8 points)

Suppose a particle is moving in a planar force field and its position vector x satisfies $\dot{x} = Ax$ and $x(0) = x_0$, where $A = \begin{bmatrix} 4 & -5 \\ -2 & 1 \end{bmatrix}$, $x_0 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find the general solution of this initial value problem.

十、(8 points)

Consider the vector space $C[-1, 1]$ with inner product defined by

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

(a) (3 points) Show that

$$u_1(x) = \frac{1}{\sqrt{2}} \quad \text{and} \quad u_2(x) = \frac{\sqrt{6}}{2}x$$

form an orthonormal set of vectors.

(b) (5 points) Use the result from part (a) to find the best least squares approximation in $\text{span}\{u_1(x), u_2(x)\}$ to $h(x) = x^{1/3} + x^{2/3}$ by a linear function.